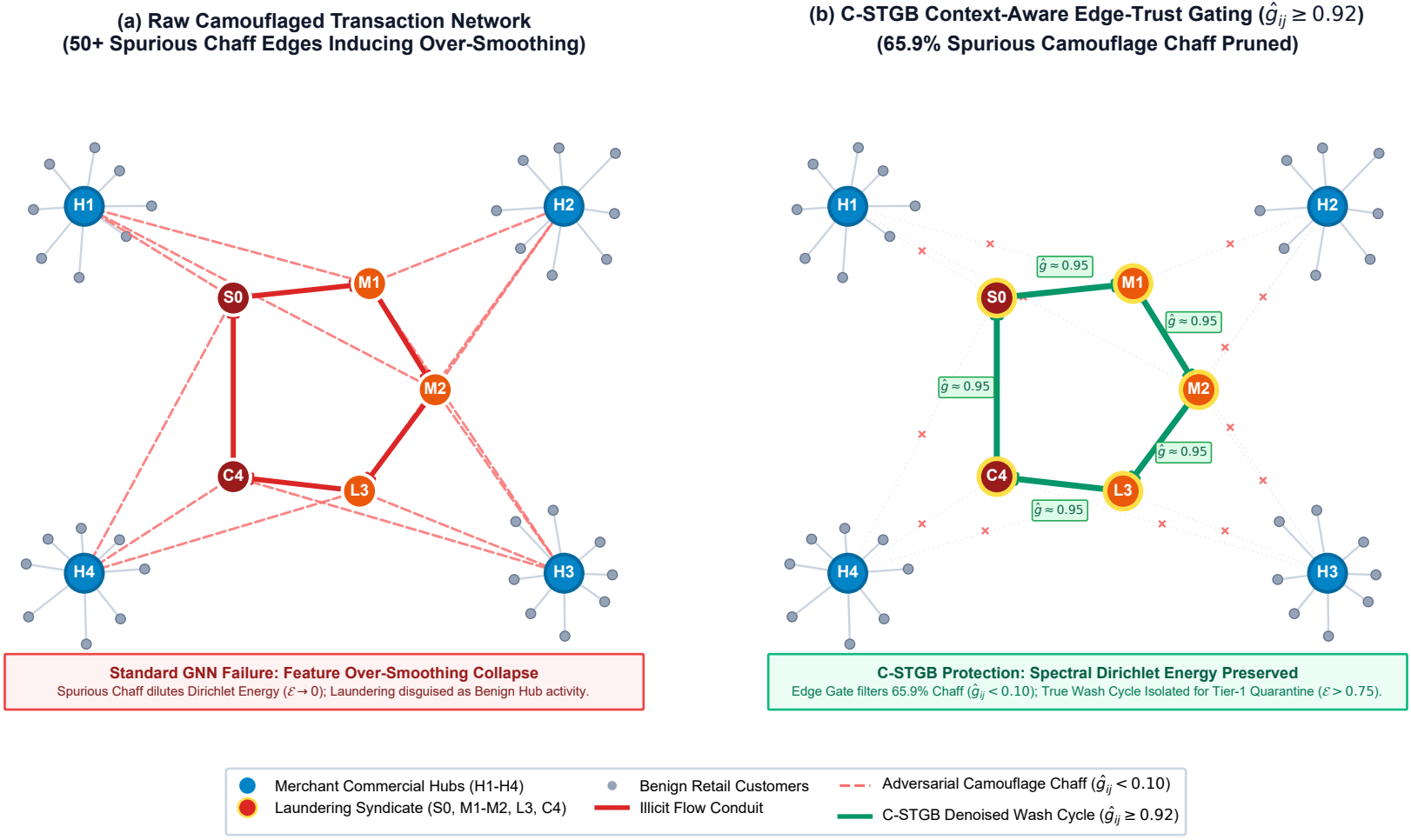

(a) Raw Camouflaged Transaction Network
(50+ Spurious Chaff Edges Inducing Over-Smoothing)
H1
H2
S0
M1
M2
C4
L3
H4
H3
Standard GNN Failure: Feature Over-Smoothing Collapse
Spurious Chaff dilutes Dirichlet Energy ($\mathcal{E} \to 0$); Laundering disguised as Benign Hub activity.
(b) C-STGB Context-Aware Edge-Trust Gating ($\hat{g}_{ij} \geq 0.92$)
(65.9% Spurious Camouflage Chaff Pruned)
H1
H2
S0
M1
M2
C4
L3
H4
H3
$\hat{g} \approx 0.95$
$\hat{g} \approx 0.95$
$\hat{g} \approx 0.95$
$\hat{g} \approx 0.95$
$\hat{g} \approx 0.95$
C-STGB Protection: Spectral Dirichlet Energy Preserved
Edge Gate filters 65.9% Chaff ($\hat{g}_{ij} < 0.10$); True Wash Cycle Isolated for Tier-1 Quarantine ($\mathcal{E} > 0.75$).
Merchant Commercial Hubs (H1-H4)
Laundering Syndicate (S0, M1-M2, L3, C4)
Benign Retail Customers
Illicit Flow Conduit
Adversarial Camouflage Chaff ($\hat{g}_{ij} < 0.10$)
C-STGB Denoised Wash Cycle ($\hat{g}_{ij} \geq 0.92$)